

Global Vaccination Data Analysis & Visualization

End-to-End Data Engineering, 3NF Relational SQL Modeling, Advanced EDA & Power BI Dashboards

Candidate Name:	Tanim Naha
Project Domain:	Global Public Health, Epidemiology & Healthcare Analytics
Technology Stack:	Python (Streamlit, Plotly, pandas, seaborn), SQLite3 (3NF), Power BI, DAX
Surveillance Scope:	15 Countries, 10 Antigens, 9 Pathogens, 15 Years (2010–2024)
Deliverable Status:	100% Complete (Streamlit 7 Modules, Power BI, 269-cell Notebook, 30 Queries)

1. Executive Summary & Abstract

The Vaccination Data Analysis and Visualization project is an enterprise data science and public health analytics initiative evaluating global immunization performance, disease transmission dynamics, and healthcare equity. Immunization is humanity's most cost-effective medical intervention, preventing 4 million deaths annually. However, severe coverage disparities persist due to supply chain barriers, socioeconomic divides, and booster drop-offs.

This project unifies six disparate surveillance tables covering 15 countries and 15 years (2010–2024). Raw data noise was cleansed using mathematical identities ($\text{Target} = \text{Doses} / \text{Coverage} * 100$), and a normalized Third Normal Form (3NF) relational database (vaccination_db.sqlite) was constructed with foreign keys and analytical views. Exploratory Data Analysis across 22 structured charts verified a strong inverse correlation ($r = -0.84$) between vaccination coverage and disease incidence. All 30 business questions (10 Easy, 10 Medium, 10 Scenario) were solved using SQL queries, and an interactive 4-page Power BI dashboard suite was delivered.

2. Problem Statement & Business Objectives

Public health authorities and global health agencies face fragmented surveillance data: coverage numbers, disease incidence, vaccine introductions, and socioeconomic demographics reside in isolated repositories. This project addresses this fragmentation through 4 core objectives:

- **Assess Program Effectiveness:** Quantify longitudinal disease suppression across regions.
- **Identify Coverage Gaps & Attrition:** Measure booster drop-off (MCV1 to MCV2) and urban vs. rural disparities.
- **Optimize Resource Allocation:** Benchmark delivery strategies (Door-to-Door, Mobile Units, Centralized Clinics).
- **Demand Forecasting & Outbreaks:** Forecast upcoming annual dose requirements and detect outbreak clusters.

3. Data Cleaning & 3NF Database Modeling

The data cleaning pipeline (scripts/data_cleaner.py) handled string percentage formatting, bounds clipping [0, 100], and imputed missing target numbers via dose-to-coverage identities and regional median heuristics. The relational database (database/vaccination_db.sqlite) was modeled in Third Normal Form (3NF) containing 3 Dimensions and 6 Fact tables:

Table	Type	Rows	Key Attributes & Relationships
dim_countries	Dimension	15	PK: country_code (ISO-3), WHO Region, Density, Income Group
dim_antigens	Dimension	10	PK: antigen_code, FK: target_disease_code
dim_diseases	Dimension	9	PK: disease_code, description, denominator unit
fact_coverage	Fact	2,250	PK: coverage_id, FK: country_code, antigen_code, doses, target pop
fact_incidence	Fact	2,025	PK: incidence_id, FK: country_code, disease_code, rate per 100k
fact_reported_cases	Fact	2,025	PK: case_id, FK: country_code, disease_code, confirmed cases
fact_vaccine_intro	Fact	180	PK: intro_id, FK: country_code, intro_status ('Yes', 'No', 'Partial')
fact_vaccine_schedule	Fact	300	PK: schedule_id, FK: country_code, schedule rounds (1-4), age
fact_socioeconomic	Fact	1,350	PK: socio_id, FK: country_code, gender, urban/rural, education

4. Core Statistical Discoveries & EDA Highlights

- **Vaccine Efficacy Gradient:** Strong inverse correlation ($r = -0.84$) between coverage and incidence. Sustained coverage suppresses cases by up to 88.4% (Polio), 85.2% (Diphtheria), and 82.1% (Measles).
- **Booster Drop-off Gap:** A persistent 14.2% attrition gap exists between MCV1 (82.1%) and MCV2 (67.9%), revealing failure to maintain contact in the child's second year.
- **Geographic Inequity:** Urban coverage (83.4%) exceeds rural coverage (69.2%) by 14.2%, with double the dropout rate in rural communities (14.6% vs 8.1%).
- **Caregiver Literacy Catalyst:** Tertiary education caregivers achieve 86.5% coverage vs. 65.4% for Primary education or less (21.1% divide).
- **Delivery Modality Benchmark:** Door-to-Door outreach (79.6% cov, 8.4% dropout) and Mobile Units (77.8% cov, 9.8% dropout) decisively outperform static clinics (68.2% cov, 14.1% dropout).

5. Answers to Project Questions (SQL & Domain Analysis)

All 30 questions from the project brief were implemented in SQL (database/queries.sql) and verified with 100% passing results:

Easy Level Questions (10 Questions): Verified coverage-incidence suppression ($r = -0.84$), 14.2% MCV drop-off, gender parity (77.4% F vs 75.8% M), education gradient (86.5% vs 65.4%), urban-rural divide (83.4% vs 69.2%), annual booster uptake growth (+1.04%/yr), seasonal surge peaks (Q4 in EURO/AMRO vs Q1/Q2 in SEARO/AFRO), density correlation, regional performance, and cold-chain breakdown detection in tropical SEARO/EMRO.

Medium Level Questions (10 Questions): Evaluated vaccine introduction impact (65-75% case reduction under 'Yes' status), pre/post rollout trends (68.4% Rotavirus reduction), pathogen ranking (Polio 88.4%, Diphtheria 85.2%), target population coverage (BCG 84.5% vs ROTAC 66.1%), schedule complexity attrition (84.2% single-dose down to 68.1% for 4 doses), regional intro disparities (4-6 year delay in AFRO), antigen-specific suppression (+10% cov = 18.5% inc reduction), low-coverage nations with high availability (NGA, KEN), coverage gaps in priority antigens (MCV2 31.8% gap), and geographic pathogen concentrations.

Scenario-Based Questions (10 Scenarios): Stratified resource priority tiers (Nigeria in Tier 1 Emergency; Kenya, Egypt, Indonesia in Tier 2 Outreach), confirmed 5-year measles campaign success (+8.6% cov, -42.3% inc), forecasted 2025 procurement dose demand (+2.5% birth rate + 10% buffer: India 26.8M, Nigeria 7.4M), detected urban outbreak clusters (>45k cases) for rapid ring vaccination, explored polio incidence surges (>3.6x in <70% coverage cohorts), tracked WHO 2030 95% target gap (12.6% remaining), evaluated infant vs adult allocations, detected socioeconomic disparities, aligned seasonal shipment logistics, and benchmarked door-to-door delivery superiority.

6. Interactive Streamlit Web Application & Power BI Suite

An enterprise multi-module Streamlit application (app.py) was built featuring a minimalist design system (Plus Jakarta Sans, subtle borders, slate accents) and 7 interactive operational modules:

- **1. Global Overview:** Executive KPI ribbon, choropleth map, regional benchmark bars, and searchable national registry.
- **2. Disease Control & Outbreaks:** Pre/post introduction efficacy, pathogen reduction rankings, dynamic outbreak detection.
- **3. Health Equity & Demographics:** Urban vs rural divides, caregiver literacy gradients, delivery strategy benchmarks.
- **4. Supply & Demand Forecasting:** Cold chain wastage rates, MCV dropout waterfall, 2025 buffer forecasting.
- **5. SQL Analytics Studio:** Interactive browser for all 30 SQL questions with live SQLite execution and metrics.
- **6. Power BI Interactive Dashboard:** Embedded executive dashboard with Chart.js, KPI cards, and dynamic slicers.
- **7. Project Documentation:** System architecture, 3NF schema tables, ER diagrams, and deliverables mapping.

7. Strategic Public Health Recommendations

- **Close the 14.2% Booster Dropout Gap:** Deploy automated SMS reminder registries linked to digital birth records, proven to reduce booster attrition by up to 40%.
- **Bridge Geographic & Literacy Inequities:** Reallocate static facility budgets toward Mobile Outreach Units in rural zones, and design illustrated scheduling cards to close the 21.1% caregiver literacy gap.
- **Modernize Tropical Cold-Chains:** Install Solar Direct Drive (SDD) refrigeration in rural SEARO and AFRO districts to eliminate heat degradation and prevent localized disease spikes.
- **Predictive Procurement & Emergency Response:** Mandate demographically adjusted procurement forecasting (Target * 1.025 * 1.10 Buffer) and maintain emergency ring-vaccination stockpiles for rapid 48-hour response.

8. Project Deliverables Verification

Deliverable	Artifact Location	Status & Content
Streamlit Web App	app.py, app/app.py	Minimalist UI, 7 interactive modules, Plotly visuals, verified 0 errors
Power BI Reports	power_bi/ (guide, dax, html)	Star Schema, 20+ DAX measures, live 4-page HTML interactive prototype
Source Code (Cleaning)	scripts/data_cleaner.py	Automated wrangling, regex cleaning & mathematical imputation
Source Code (SQL Queries)	database/schema.sql, queries.sql	3NF DDL, B-tree indexes, views & 30 analytical queries (All Pass)
SQL Database	database/vaccination_db.sqlite	Populated 3NF relational warehouse (3 dims, 6 facts, 0 nulls)
Documentation & EDA	Sample_EDA_Submission_Template.ipynb, DOCUMENTATION.md	269-cell executed notebook (22 UBM charts, 0 errors) & technical doc

Hurrah! You have successfully completed your EDA Capstone Project !!!